Predicting Provincial Rice Yield in the Philippines from Fertilizer, Climate, and Soil Conditions: A Survey-Constrained Panel Study

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***Abstract*—Province-level rice yield in the Philippines depends on fertilizer, climate, and soil, but the resolution at which this can be modeled is set by the coarsest public data source. We predict average rice yield (metric tons per hectare) from 287 province-season-ecosystem observations covering 76 of about 81 reporting provinces, two season-years, and irrigated and non-irrigated ecosystems; soil pH is a simulated proxy, not a measurement. Ridge Regression, Random Forest, and LightGBM were compared against a Linear Regression baseline under a province-grouped 80/20 split with leave-one-province-out cross-validation, with stratified mixup and Gaussian copula augmentation also tested. The selected model, Ridge Regression with 3× mixup, reached a validation RMSE of 0.667 (R^2 0.387) and, in one evaluation on a sealed 16-province test set, RMSE 0.702 (R^2 0.282). Post hoc baselines show how little of this comes from fertilizer, climate, and soil: a lookup of the training mean for each season, ecosystem, and island-group cell reaches RMSE 0.704 in validation and 0.619 on the test set, and its difference from the selected model is not statistically distinguishable in either. Ecosystem and island group carry most of the predictable variation. Mixup showed no reliable benefit across random seeds, and four rows from one province account for 63% of test squared error. The results are consistent with an information ceiling set by feature granularity. The selected model was deployed as a local Streamlit application.**

***Index Terms*—rice yield prediction, fertilizer application, spatial cross-validation, data augmentation, regression**

I. Introduction

Rice is the Philippines' staple crop, and provincial yield depends on interacting agronomic factors: fertilizer application, climate, and soil chemistry. Machine learning has been widely applied to crop yield prediction, with neural networks, linear regression, and random forests among the most frequently used methods, although frequency of use is not evidence of accuracy [1]. Practical nutrient guidance for rice in Asia has instead relied on field-specific, rule-based approaches such as site-specific nutrient management [2] and software tools such as Nutrient Manager that turn answers to a few field questions into nitrogen, phosphorus, and potassium recommendations [3]. We did not locate a published study that learns province-level rice yield jointly from Philippine fertilizer application-rate statistics, climate, and soil, which motivates a supervised regression approach for coarse, illustrative decision support.

The research objective is to determine how well three traditional machine learning algorithms can predict province-level rice yield under the real constraints of public Philippine data, and to state honestly where those constraints, rather than the modeling choice, limit accuracy. The intended user is an agricultural extension worker or analyst who needs a coarse province-level estimate, not a farm-level production tool. The scope is set by data availability: 287 province-ecosystem-season observations across two season-years (2021 Semester 2 and 2022 Semester 1) and 76 provinces, as discussed in Section IV.

II. Methods

A. Dataset and Ethics

All data sources are public. Yield, area harvested, and fertilizer application-rate statistics come from the Philippine Statistics Authority (PSA) through the PhilRice Ricelytics portal [4]. Climate data (rainfall, temperature, solar radiation) come from the Open-Meteo historical weather archive [5], aggregated per semester at a single point per province (its centroid). No personally identifiable or sensitive data are involved.

Soil pH is a simulated feature, not a measurement: a fixed-seed draw per province and ecosystem (5.6–6.5 irrigated, 5.0–6.1 non-irrigated). It was simulated because we found no public province-representative soil pH panel. The ranges are our assumptions, chosen to be broadly consistent with reported Philippine paddy-soil pH, which is slightly to strongly acid across major irrigated areas [6] and, in one provincial survey, about 5.4–6.3 for irrigated and 4.0–7.1 for rainfed soils [7]; they are not taken from a single source. Because the ranges differ by ecosystem, simulated pH partly encodes ecosystem, and Sections III-A and III-F quantify the consequence. Fertilizer source tables report kilograms without an explicit per-hectare denominator; a per-hectare reading is assumed from plausible agronomic magnitude and is documented as an assumption, not a confirmed unit.

B. Features and Target

The target is `yield_mt_ha`, average palay yield in metric tons per hectare (mean 3.61, median 3.59, SD 0.85, range 1.22–6.77). The 14 features are three categorical (`season_period`, `ecosystem`, `island_group`) and eleven numeric (area harvested, N/P/K application rates, total

fertilizer intensity, a fertilizer diversification index, rainfall, solar radiation, soil pH, mean temperature, and diurnal temperature range). Province identity was kept only as a grouping identifier for validation, never as a trained feature. The full data dictionary is in Appendix A.

Five individual fertilizer grade columns were dropped because they are redundant with their own sum (urea–total correlation 0.818) and replaced by total intensity and the diversification index. Raw maximum and minimum temperature (correlation 0.848) were collapsed into mean and range. A fertilizer grade absent from a province's top-five list is encoded 0, meaning "unlisted," not confirmed zero use; rows with an incomplete element-rate record were excluded rather than zero-filled.

C. Exploratory Data Analysis Procedure

EDA covered data integrity (missing values, duplicate rows, duplicate province-season-ecosystem keys, and consistency of derived columns), the target distribution, the effect of ecosystem and season, outliers by the 1.5×IQR rule, feature skewness, pairwise correlations with yield, and collinearity among features. Correlations were computed on all rows and, after the first model evaluation, within each ecosystem. Findings are reported in Section III-A.

D. Preprocessing

A `ColumnTransformer` applied standard scaling to numeric features (area harvested was log-transformed first) and one-hot encoding with the first category dropped to the three categorical features, avoiding the dummy-variable trap under linear solvers. The encoded matrix has 15 columns. All preprocessing was fit on training rows only and refit within every cross-validation fold to prevent leakage. Implementation used scikit-learn [8].

E. Split Strategy

A province-grouped 80/20 split (`GroupShuffleSplit`, seed 42) placed 229 rows from 60 provinces in training and 58 rows from 16 provinces (Abra, Albay, Benguet, Bohol, Camiguin, Dinagat Islands, Ilocos Sur, Isabela, Misamis Oriental, Negros Occidental, Northern Samar, Occidental Mindoro, Quezon, South Cotabato, Sultan Kudarat, Tarlac) in a sealed test set, with zero province overlap, confirmed directly. Within the training provinces, models were compared with leave-one-province-out cross-validation (60 folds) instead of standard k-fold, because ecosystem pairs from one province share near-identical fertilizer, climate, and soil values and random folding would leak between related rows [9]. Preprocessing and augmentation were refit inside each fold using that fold's training rows only.

F. Algorithms and Tuning

Four models were trained under identical conditions: Linear Regression (reported baseline, not eligible for selection) and the three compared algorithms, Ridge Regression ($\alpha = 1.0$) [10], Random Forest (500 trees, minimum leaf size 2) [11], and LightGBM [12]. With about 225 training rows per fold, LightGBM was deliberately constrained (200 estimators, learning rate 0.05, 7 leaves, depth 3, minimum 25 samples per leaf, column sampling 0.8, ℓ_1 penalty 1.0, ℓ_2 penalty 5.0) to limit its leaf-wise overfitting.

No hyperparameter search was run. All settings above were fixed by hand a priori for the small-sample regime, because a nested search over 60 groups of 2–4 rows would itself be unstable. The comparison is therefore between reasonable fixed configurations, not tuned optima (Section IV-E).

G. Data Augmentation

Training folds were optionally augmented with stratified mixup [13]: synthetic rows are convex combinations ($\lambda \sim \text{Beta}(0.4, 0.4)$) of pairs of real rows sharing the same season, ecosystem, and island group, so blended rows stay agronomically coherent. A multiplier m produces $(m-1)n$ synthetic rows from n real rows ($1\times$ is the no-augmentation control); multipliers 1, 2, 3, and 5 were compared by leave-one-province-out validation, and the multiplier with the lowest cross-model mean RMSE was selected. Real and synthetic rows were weighted 1.0 and 0.5. Each configuration used one random seed (42 offset by fold index); seed sensitivity was examined afterwards (Section III-G). Synthetic yields at $2\times$ stayed inside the real range ([1.37, 6.74] versus [1.22, 6.77]), as expected of interpolation, and no synthetic row had an impossible categorical combination.

As a contrast, a Gaussian copula generator (sdv 1.38.5 [14]; normal marginals, fit per fold on raw training rows, values clipped to fold range) was also tested. Its fidelity was confirmed (fertilizer–yield correlation 0.376 synthetic versus 0.347 real; pH–yield 0.393 versus 0.412). Synthetic data never entered validation or test rows.

H. Evaluation Metrics and Post Hoc Analyses

RMSE is the primary metric because it penalizes large provincial errors more than MAE; MAE, R^2, and residuals are supporting metrics. Validation uses pooled out-of-fold predictions, because individual folds hold only 2–4 rows. Uncertainty was estimated by bootstrap [15] over rows and, because rows from one province are correlated, over provinces. The test set was evaluated once, on the selected model only, after selection from validation evidence alone. Simple baselines, feature-group ablations, and seed checks (Sections III-F and III-G) were run afterwards, are reported as post hoc context, and did not influence selection.

III. Results

A. EDA Findings

The merged dataset has zero missing values, zero duplicate rows, zero duplicate province-season-ecosystem keys, and a consistent derived feature (total fertilizer equals the sum of its five components to about 5.7×10^{-14}). The target is nearly symmetric (skewness 0.06), so no transformation was applied. Irrigated yields are clearly higher than non-irrigated yields (median 4.08 versus 3.12 MT/ha; Fig. 1); season differs little

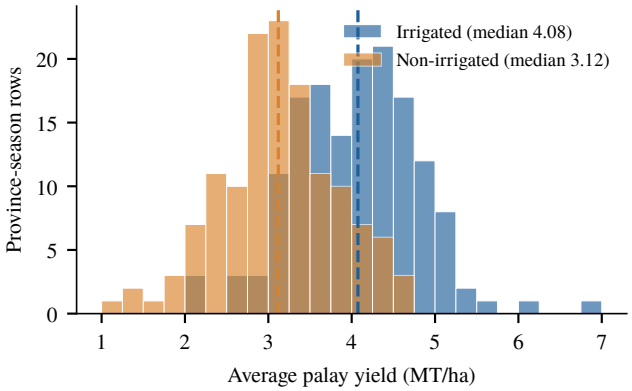


Fig. 1. Distribution of average yield by ecosystem (287 rows). Dashed lines mark medians.

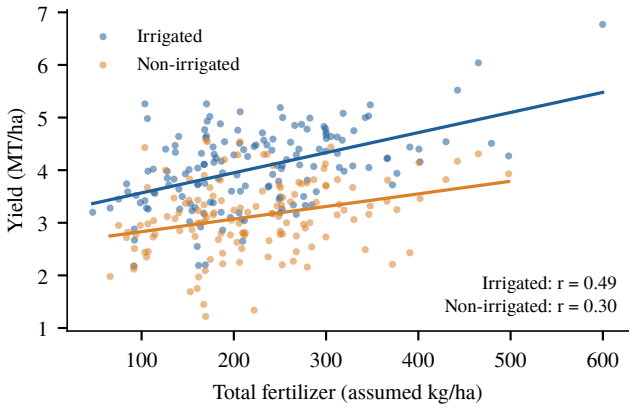


Fig. 2. Yield against total fertilizer by ecosystem with within-ecosystem fits. The relationship is positive in both ecosystems but leaves most variation unexplained.

(medians about 3.6 versus 3.5), and the two season periods are also two different calendar years, so season and year cannot be separated.

Two observations lie outside the 1.5×IQR bounds ([1.23, 6.05]) and were retained as legitimate extremes: Davao Occidental (non-irrigated, 2021 S2, yield 1.22, fertilizer 169.4 kg/ha) and Nueva Ecija (irrigated, 2022 S1, yield 6.77, the dataset maximum fertilizer 599.8 kg/ha), both internally consistent. Area harvested is right-skewed (3.158) and was log-transformed; its irrigated mean (21,746 ha) exceeds the non-irrigated (9,850 ha), so it partly proxies ecosystem and scale and is not interpreted causally.

Correlations with yield are modest: area harvested 0.437, soil pH 0.380, total fertilizer 0.353, nitrogen rate 0.335, and rainfall −0.112 (weakest). Two of the top four need care. Simulated pH correlates 0.69 with irrigated status, and within each ecosystem its correlation with yield is −0.02 (irrigated) and +0.04 (non-irrigated), so its 0.380 is inherited from ecosystem by construction and is not evidence of a soil effect. Total fertilizer keeps a positive relationship within each ecosystem (Fig. 2; $r = 0.49$ irrigated, 0.30 non-irrigated), whereas area harvested correlates 0.50 with yield among irrigated rows but only 0.13 among non-irrigated rows.

B. Model Comparison

Table I compares all four models without augmentation and at 3× mixup (all 28 model-condition results are in Appendix B). Ridge with 3× mixup has the lowest RMSE of all (0.6669), with Linear Regression a near-tie (0.6681) that is ineligible for selection. Random Forest has the best MAE in the no-augmentation control (0.5334) and at 3× (0.5385) but a worse RMSE than Ridge at every multiplier, indicating smaller typical errors with occasional larger misses. LightGBM has the worst RMSE of the three compared algorithms at 1× and 3×, plausibly because its flexibility is poorly matched to about 225 rows despite regularization.

The cross-model mean RMSE across the mixup ladder was 0.6826 (1×), 0.6873 (2×), 0.6775 (3×), and 0.6905 (5×), so

TABLE I
VALIDATION RESULTS, LEAVE-ONE-PROVINCE-OUT, POOLED OUT-OF-FOLD

Condition	Model	RMSE	MAE	R^2
No augmentation	Linear (baseline)	0.6772	0.5511	0.3673
	Ridge	0.6749	0.5488	0.3716
	Random Forest	0.6758	0.5334	0.3700
	LightGBM	0.7027	0.5589	0.3188
Mixup 3×	Linear (baseline)	0.6681	0.5414	0.3842
	Ridge (selected)	**0.6669**	**0.5405**	**0.3865**
	Random Forest	0.6838	0.5385	0.3548
	LightGBM	0.6912	0.5443	0.3409

3× was selected by the lowest-mean-RMSE rule; the spread is small and, as Section III-G shows, comparable to seed variation. Copula augmentation was a validated negative result: cross-model mean RMSE was 0.6936 (2×), 0.6960 (3×), and 0.7010 (5×), worse than mixup at the same multiplier and than no augmentation. We attribute this, plausibly but untested, to the difficulty of estimating a 15-dimensional dependence structure (105 pairwise correlations) from about 225 rows.

C. Uncertainty and Final Selection

Row-level 95% bootstrap intervals for RMSE (2000 resamples) overlap substantially: Ridge with 3× mixup 0.6669 [0.6079, 0.7184]; Ridge without augmentation 0.6749 [0.6150, 0.7280]; Ridge with copula 3× 0.6844 [0.6231, 0.7377]; Random Forest with 3× mixup 0.6838 [0.6188, 0.7437]. The province-level bootstrap interval for the selected configuration is wider, [0.589, 0.740], as expected when rows within a province are correlated; the per-fold RMSE standard deviation is 0.29. Ridge with 3× mixup was selected before any test access for the lowest pooled RMSE among the three compared algorithms, interpretable coefficients, and a smaller RMSE–MAE gap than Random Forest. The margins between algorithms are small relative to these intervals, so the choice is defensible but not statistically decisive.

D. Final Test Evaluation

On the sealed 58-row, 16-province test set, evaluated once, the selected model achieved RMSE 0.7022, MAE 0.4747, and R^2 0.2818. The test RMSE lies inside the validation intervals (row-level [0.608, 0.718], province-level [0.589, 0.740]), consistent with leave-one-province-out expectations and showing no evidence of validation overfitting, though with only 16 test provinces the estimate is itself imprecise. Test MAE is lower than validation MAE (0.475 versus 0.541) while RMSE and R^2 are worse: a few large errors inflate squared-error metrics but barely move MAE, and the test target is slightly less variable (SD 0.829 versus 0.851), which lowers R^2 for the same error. The typical miss of about 0.70 MT/ha is roughly 19% of the mean yield of 3.61.

E. Residual Analysis

Residuals show no funnel shape or curvature (Fig. 3b), but error is highly concentrated: four rows from one province, Dinagat Islands (actual 1.97–2.20 MT/ha, predicted 3.66–4.73), account for 63% of test squared error, and excluding that province RMSE on the other 54 rows is 0.4447. This is a diagnostic, not a replacement for the headline result, and it is not like-for-like with validation, which contains its own hardest provinces. The mean test residual is −0.30 MT/ha (−0.17 without Dinagat), so the model over-predicts on these provinces on average.

The data suggest why Dinagat is missed. Its irrigated-minus-non-irrigated yield gap is 0.11 and 0.22 MT/ha in the two seasons against a dataset median of 0.75 (the 6th and 13th smallest of 135 province-seasons), so its irrigated rows behave like rainfed rows, against the ecosystem pattern the model relies on. Its phosphorus and potassium rates rank among the three highest of 76 provinces while total fertilizer is below average (169 and 162 versus means of 234 and 215 kg/ha). The model read the high P and K as higher yield, adding roughly 0.5–0.7 MT/ha relative to the average training row in its ecosystem-island-season cell, while observed yields were among the lowest in the country. A smaller miss occurred at Negros Occidental (non-irrigated, 2022 S1: actual 1.45, predicted 2.83). Mean absolute error is near parity across ecosystems (irrigated 0.459, non-irrigated 0.494).

F. Comparison with Simple Baselines (Post Hoc)

After the test evaluation we asked how much of the accuracy comes from fertilizer, climate, and soil rather than from where and how rice is grown. Three fixed predictors were scored under the same protocols using training provinces only: the training mean; the training mean per ecosystem; and the training mean per season × ecosystem × island-group cell (12 cells, at least 12 training rows each). The selected model was not changed on this basis.

Against the training mean the selected model's RMSE is 22% lower in validation and 17% lower on test; against the ecosystem mean, 8% and 3% lower; against the 12-cell lookup, 5% lower in validation but 13% higher on test (test R^2 0.442 for the lookup versus 0.282). The reversal is produced by

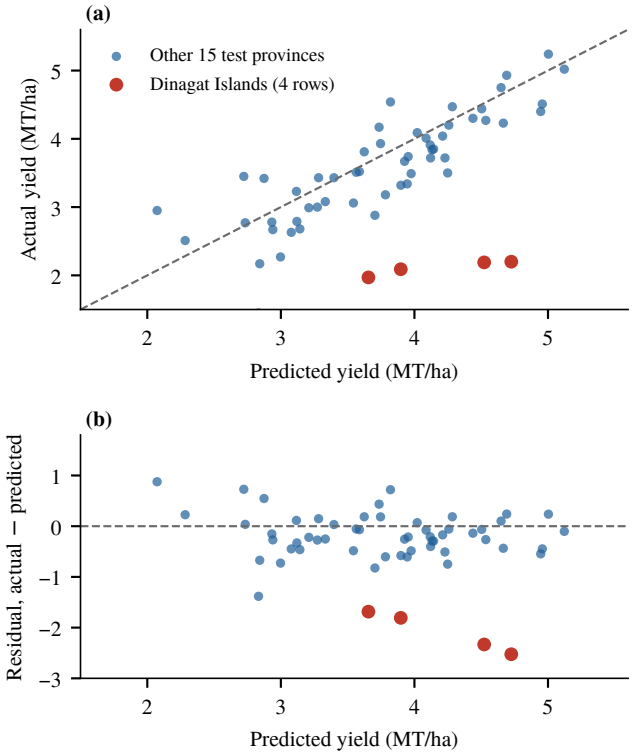


Fig. 3. Sealed test set. (a) Actual versus predicted yield; the dashed line is perfect prediction. (b) Residuals versus predicted yield. The four Dinagat Islands rows (red) account for 63% of squared error.

TABLE II
SELECTED MODEL VERSUS SIMPLE BASELINES (RMSE, MT/HA)

Predictor	Validation	Test
Training mean	0.859	0.845
Mean per ecosystem	0.725	0.722
Mean per season×ecosystem×island	0.704	0.619
Selected: Ridge, mixup 3×	**0.667**	**0.702**

TABLE III
RIDGE MINUS 12-CELL LOOKUP: PAIRED RMSE DIFFERENCE (PROVINCE-LEVEL BOOTSTRAP)

Evaluation	ΔRMSE	95% CI	P(Ridge better)
Validation (60 prov.)	−0.037	[−0.128, 0.042]	0.80
Test (16 prov.)	+0.083	[−0.111, 0.236]	0.29
Pooled (76 prov.)	−0.014	[−0.100, 0.065]	0.64

Dinagat: on its four rows Ridge's squared error is 17.92 against 9.69 for the lookup, while on the other 54 rows Ridge is better (10.68 versus 12.53; RMSE 0.445 versus 0.482), and Ridge is better in 9 of 16 test provinces. A province-level paired bootstrap (Table III) shows the differences are not distinguishable from zero in validation, on test, or pooled across all 76 provinces.

The saved Ridge coefficients agree: they are dominated by

TABLE IV
RIDGE FEATURE-GROUP ABLATIONS (VALIDATION, NO AUGMENTATION)

Features used	RMSE	R^2
All 14 features	0.675	0.372
Without soil pH (simulated)	0.669	0.382
Without climate (4 variables)	0.663	0.393
Without fertilizer (5 variables)	0.694	0.336
Season, ecosystem, island, area only	0.685	0.352
Season, ecosystem, island only	0.707	0.311
Ecosystem only	0.725	0.276

non-irrigated ecosystem (-0.85 MT/ha) and Visayas (-0.54), and every numeric feature contributes 0.14 MT/ha or less per standard deviation. Feature-group ablations (Table IV) show fertilizer is the only group that clearly helps; removing simulated pH or climate slightly lowers validation RMSE. Note that adding area harvested to the three categorical features improves RMSE by about as much (0.022) as fertilizer does (0.019), and area is largely an ecosystem and scale proxy. The ablations carry no confidence intervals, so we report them as diagnostics only.

G. Seed Sensitivity of Mixup (Post Hoc)

The validation sweep used a single seed per configuration. Re-running Ridge with mixup over 11 seeds (Appendix D) gives mean RMSE 0.6770 (SD 0.0049) at $3\times$, 0.6784 at $2\times$, and 0.6778 at $5\times$, against 0.6749 without augmentation. The reported 0.6669 is the best of the 11 seeds at $3\times$, so mixup showed no reliable benefit for Ridge and the $3\times$ result is best read as a favorable random draw.

H. Application Evidence

The selected model and preprocessing pipeline are deployed as a local Streamlit application that accepts season, ecosystem, island group, area harvested, soil pH, climate, and per-element and per-grade fertilizer inputs (province is not an input, consistent with its exclusion as a feature). Eight scenarios were run (sample inputs and outputs in Appendix C). The criterion was that the application returned a finite prediction without error on valid, boundary, and logically inconsistent inputs, including minimum temperature above maximum, which the interface does not validate. Several scenarios lie outside the training range (zero fertilizer, 10 ha, 3500 mm), so they test robustness, not prediction validity. Predictions moved in a plausible direction for ecosystem, area, and fertilizer; yield also rose with rainfall although the raw correlation is weakly negative (-0.112), which should not be read as an agronomic effect.

IV. DISCUSSION

A. Why Ridge Performed Best, and What That Means

Ridge was best on RMSE, but all four models fall within about 0.03 RMSE of each other, within bootstrap noise, and none clearly beats a lookup that uses only season, ecosystem, and island group (Table II). The honest reading is that the model's skill comes mostly from two facts, whether the land is irrigated and which island group it is in, with fertilizer adding a small signal and climate and simulated pH adding essentially none. Against the training mean the model cuts RMSE by 22% (validation) and 17% (test), but that is the weakest baseline; against the ecosystem mean the reductions are 8% and 3%.

B. An Information Ceiling, Not a Modeling Failure

The evidence points to the data rather than the algorithm. (i) The 12-cell lookup reaches validation R^2 0.32 against the model's 0.39 and is statistically tied with it. (ii) Only fertilizer helps in the ablations. (iii) Linear and tree models converge to similar RMSE, evidence against substantial unexploited nonlinear structure. (iv) Random Forest and LightGBM fit training rows far better than held-out provinces (training R^2 0.91 and 0.79 versus validation 0.37 and 0.32; Appendix D), so the limit is generalizable signal, not an under-fit model. (v) In an exploratory learning curve, validation RMSE flattens between 75% and 100% of training provinces. Two cautions apply: convergence across models is also consistent with 225 rows being too few for trees to find structure, and the learning curve used one or two repeats. The results are therefore consistent with a ceiling set by feature granularity, not proof of one. The feature set is coarse by construction: province-mean fertilizer, season-aggregated climate at one point per province, and simulated pH.

C. A Granularity Ceiling, Not a Sampling Limit

The study size follows from the publication window of the fertilizer application-rate survey, available only for 2021 Semester 2 and 2022 Semester 1. Yield, area, and climate sources extend across longer periods; the binding constraint is their intersection with fertilizer data. Seventy-six of about 81 reporting provinces are represented, closer to a near-census of published provinces than a small sample. That claim concerns provinces, not time: with two seasons, the data say little about year-to-year weather variation. Province-season-ecosystem is the finest resolution at which fertilizer application rates are published, so the limitation is framed as a granularity ceiling, addressed here by grouped validation, province-level uncertainty, and seed checks rather than by unsupported data expansion.

D. Mixup, Errors, and Tradeoffs

Mixup effects were small and inconsistent: relative to no augmentation, Ridge improved only at $3\times$ (-0.008) and was worse at $2\times$ and $5\times$; Random Forest was worse at every multiplier; LightGBM improved at every multiplier by about 0.012; all within seed noise (Section III-G). It is reported as a null result alongside the copula negative result. The main error tradeoff is that the model trades a small gain in typical accuracy for large misses on provinces whose irrigation or fertilizer profile departs from the pattern (Dinagat Islands, Negros Occidental); RMSE was chosen as the primary metric precisely because such misses are costly for decisions.

E. Influential Observations and Limitations

Nueva Ecija (highest yield and fertilizer in the dataset) and Dinagat Islands (dominant test error) are single provinces with outsized influence on their evaluation splits, a structural consequence of grouped evaluation at this size, not an error to be removed. Limitations are:

- the granularity ceiling, and only two seasons that are also two years;
- simulated soil pH whose range depends on ecosystem, giving no independent soil evidence;
- an assumed (unconfirmed) per-hectare fertilizer unit and a top-five grade encoding that does not mean zero use;
- climate taken at one point per province, and area harvested, which is known only after harvest, so the model is a retrospective estimate, not a forecast;
- no hyperparameter search and one seed per configuration in the validation sweep;
- a 16-province test set dominated by one province, and baselines added after the test was opened;
- overlapping confidence intervals across augmentation settings and algorithms;
- observational data, so coefficients are associations, not causal effects;
- deployment tests mostly outside the training range, no temperature validity check, and local-only deployment.

V. Conclusion and Recommendations

Province-level rice yield can be predicted with modest skill from public data, and most of that skill comes from knowing whether the land is irrigated and which island group the province belongs to. Fertilizer adds a small signal within ecosystems; climate and the simulated soil pH add little. The selected model, Ridge Regression with $3\times$ stratified mixup, reached a one-time test RMSE of 0.702 (MAE 0.475), statistically consistent with validation and not distinguishable from a simple 12-cell lookup, with test error concentrated in one atypical province. Mixup showed no reliable benefit. Its practical value is as a coarse, illustrative, retrospective estimate, not a farm-level or causal tool.

Realistic improvements are municipal-level yield paired with gridded rather than centroid climate and soil estimates (stating any carried-forward fertilizer assumptions explicitly), measured soil data in place of the simulated proxy, more seasons to separate season from year, hyperparameter search with province-grouped nested validation, and a naive-baseline comparison planned before the test set is opened. A more ambitious direction outside this course project is variety-specific nitrogen-potassium response modeling with lodging and pest or disease risk constraints.

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Appendix A
Data Dictionary

One row is a province-season-ecosystem observation (287 rows, 76 provinces, 2 season periods $\times$ 2 ecosystems). `province` is kept for auditability and leave-one-province-out grouping only; `island_group` is the geographic feature. By-element source headers report `kg` without a per-hectare denominator; reading them as `kg/ha` is an assumption based on plausible magnitudes. `soil_ph` is a fixed-seed simulated proxy (SHA-256 seed of province|ecosystem; uniform 5.6–6.5 irrigated, 5.0–6.1 non-irrigated; one value per province-ecosystem, constant across season years), reproducible and not a measurement. A grade absent from a province's top-five list is encoded 0 meaning "unlisted."

Final feature matrix. 14 trained features: 3 categorical + 11 numeric (`area_harvested_ha`, `n/p/k_application_rate`, `total_fertilizer_kg_ha`, `fertilizer_diversification_index`, `rainfall_mm`, `solar_radiation_mj_m2`, `soil_ph`, `temp_mean_c`, `temp_range_c`).

TABLE V
DATA DICTIONARY: RAW MERGED COLUMNS AND MODELING ROLE

Column	Description / Source	Unit or Type	Role
season_period	PSA yield period matched to the fertilizer season (2021_Semester_2, 2022_Semester_1).	category	Trained feature (one-hot)
ecosystem	PSA rice ecosystem (Irrigated / Non-irrigated).	category	Trained feature (one-hot)
province	Province name, canonicalized via aliases.	identifier	Grouping/audit only; never a trained feature
island_group	PSGC island group (Luzon / Visayas / Mindanao).	category	Trained feature (one-hot); geographic proxy
yield_mt_ha	Average palay yield, PSA.	MT/ha	TARGET
area_harvested_ha	Rice area harvested, PSA.	ha	Trained; log1p-transformed (skew 3.16); scale/management proxy, not causal
n_application_rate	Nitrogen application rate.	assumed kg/ha	Trained feature
p_application_rate	Phosphorus application rate.	assumed kg/ha	Trained feature
k_application_rate	Potassium application rate.	assumed kg/ha	Trained feature
grade_ammophos_16_20_0_kg_ha, grade_ammosul_21_0_0_kg_ha, grade_complete_14_14_14_kg_ha, grade_mop_0_0_60_kg_ha, grade_urea_46_0_0_kg_ha	Reported top-five grade quantities; absent means unlisted, not proven zero use.	assumed kg/ha	Source-only: excluded from the feature matrix (redundant with total; urea vs. total $r = 0.818$); kept for audit, total/entropy, and app inputs
total_fertilizer_kg_ha	Sum of the five listed grade columns (verified; max abs. diff $\sim 5.7 \times 10^{-14}$).	assumed kg/ha	Trained feature
fertilizer_diversification_index	Shannon entropy of shares among the five listed grades.	dimensionless	Trained feature
rainfall_mm	Semester total daily precipitation, Open-Meteo.	mm	Trained feature
temp_max_c	Mean daily maximum temperature over the semester, Open-Meteo.	°C	Source-only: derives temp_mean_c/temp_range_c ($r = 0.848$ vs. temp_min)
temp_min_c	Mean daily minimum temperature over the semester, Open-Meteo.	°C	Source-only (see temp_max_c)
solar_radiation_mj_m2	Mean daily shortwave radiation over the semester, Open-Meteo.	MJ/m^2/day	Trained feature
soil_ph	Fixed-seed simulated proxy per province/ecosystem; not measured.	simulated pH	Trained feature

TABLE VI
DATA DICTIONARY: DERIVED FEATURES

Column	Definition	Rationale
temp_mean_c	(temp_max_c + temp_min_c) / 2	Average thermal regime
temp_range_c	temp_max_c − temp_min_c	Diurnal swing; replaces the collinear raw bounds

Pipeline (fit on training only): log1p then standard scaling on area; standard scaling on the other 10 numerics; `OneHotEncoder(drop='first', handle_unknown='ignore')` on the 3 categoricals, giving 15 columns (order in `artifacts/feature_order.json`). Excluded from training: province, the five grade columns, raw temperature bounds, and all 58 test rows.

APPENDIX B
FULL AUGMENTATION RESULTS

TABLE VII
MIXUP: FULL RESULTS (POOLED OOF, 60 FOLDS, SEED 42)

Mult.	Model	RMSE	MAE	R^2
1×	Linear	0.6772	0.5511	0.3673
1×	Ridge	0.6749	0.5488	0.3716
1×	Random Forest	0.6758	0.5334	0.3700
1×	LightGBM	0.7027	0.5589	0.3188
2×	Linear	0.6816	0.5552	0.3591
2×	Ridge	0.6798	0.5533	0.3625
2×	Random Forest	0.6979	0.5510	0.3281
2×	LightGBM	0.6900	0.5414	0.3433
3×	Linear	0.6681	0.5414	0.3842
3×	**Ridge (selected)**	**0.6669**	**0.5405**	**0.3865**
3×	Random Forest	0.6838	0.5385	0.3548
3×	LightGBM	0.6912	0.5443	0.3409
5×	Linear	0.6837	0.5550	0.3552
5×	Ridge	0.6827	0.5542	0.3570
5×	Random Forest	0.7053	0.5518	0.3138
5×	LightGBM	0.6905	0.5429	0.3422

TABLE VIII
GAUSSIAN COPULA: FULL RESULTS (POOLED OOF, 60 FOLDS)

Mult.	Model	RMSE	MAE	R^2
$2\times$	Linear	0.6771	0.5443	0.3674
$2\times$	Ridge	0.6770	0.5436	0.3676
$2\times$	Random Forest	0.7091	0.5586	0.3063
$2\times$	LightGBM	0.7109	0.5598	0.3027
$3\times$	Linear	0.6844	0.5479	0.3538
$3\times$	Ridge	0.6844	0.5475	0.3539
$3\times$	Random Forest	0.7023	0.5527	0.3196
$3\times$	LightGBM	0.7129	0.5652	0.2988
$5\times$	Linear	0.6950	0.5569	0.3337
$5\times$	Ridge	0.6949	0.5566	0.3337
$5\times$	Random Forest	0.7054	0.5525	0.3135
$5\times$	LightGBM	0.7086	0.5659	0.3073

APPENDIX C
SAMPLE INPUTS AND OUTPUTS

TABLE IX

DEPLOYMENT TEST SCENARIOS. PREDICTIONS REPLAYED WITH THE DEPLOYED MODEL; UNLISTED INPUTS AT APPLICATION DEFAULTS (AREA 5,000 HA, PH 5.8, RAINFALL 1500 MM, SOLAR RADIATION 17.5 MJ/M^2/DAY, T_{max} 30 °C, T_{min} 23 °C, N/P/K 75/15/15, GRADES UREA 80, COMPLETE 70, AMMOPHOS 15, AMMOSUL 10, MOP 5, LUZON, IRRIGATED, 2021 S2). TRAINING-DATA RANGES: TOTAL FERTILIZER 47–600, AREA 20–153,468 HA, RAINFALL UP TO 3,239 MM.

Scenario (changed input)	Yield (MT/ha)	Note
Baseline (Luzon, irrigated, defaults)	4.19	Reference for the rows below
Zero fertilizer input	3.58	Below training minimum; no division by zero in diversification index
Single grade only (urea 200 kg/ha)	4.68	Diversification index 0
Extreme area (150,000 ha)	4.48	Near the data maximum; log1p handles the tail
Minimal area (10 ha), Visayas, non-irrigated	2.27	Below the data minimum of 20 ha
Extreme rainfall (3500 mm)	4.39	Above the data maximum; no overflow
$T_{min} > T_{max}$ (invalid physical input)	3.17	No crash; interface lacks a validity check
Mindanao, non-irrigated, 2022 S1	3.23	In range

TABLE X

RIDGE VALIDATION RMSE ACROSS MIXUP SEEDS (11 SEEDS)

Setting	Mean	SD	Min	Max
No augmentation (1×)	0.6749	–	–	–
Mixup 2×	0.6784	0.0031	0.6757	0.6840
Mixup 3×	0.6770	0.0049	0.6669	0.6830
Mixup 5×	0.6778	0.0047	0.6701	0.6854

TABLE XI

TRAINING VERSUS VALIDATION FIT (1×, NO AUGMENTATION)

Model	Training R^2	Validation R^2
Ridge	0.540	0.372
Random Forest	0.910	0.370
LightGBM	0.785	0.319

TABLE XII

LEARNING CURVE: VALIDATION RMSE USING A FRACTION OF THE TRAINING PROVINCES (EXPLORATORY)

Fraction of training provinces	25%	50%	75%	100%
Ridge (mean of 2 repeats)	0.838	0.732	0.670	0.675
Random Forest, 200 trees (1 repeat)	0.711	0.704	0.678	0.675

APPENDIX D
SUPPLEMENTARY DIAGNOSTICS (POST HOC)

These checks were run after the one-time test evaluation, are reported for context, and did not influence model selection. They are reproduced by `src/diagnostics/extra_checks.py`.

APPENDIX E
REPRODUCIBILITY LOG

The following issues were found and resolved during development:

1) A custom `FunctionTransformer` lacked `get_feature_names_out`; fixed with `feature_names_out="one-to-one"`.
2) The synthetic-data library's valid marginal identifier is `'norm'`, not `'gaussian'`.
3) A 5-province smoke-test run of the copula pipeline was briefly merged into the main metrics and misread as a positive result; caught by the fold-count column, and smoke-test rows are no longer interpreted as results.
4) A metrics merge on re-run overwrote technique labels and duplicated rows; `model_metrics.csv` (28 rows) was rebuilt from `oof_predictions.csv` as the single source of truth and the merge made idempotent.
5) A final-fit cell was briefly run against the copula configuration instead of the selected mixup configuration; caught through an unexpected warning and corrected.
6) Notebooks do not share memory, so each reloads the dataset and split files; saved artifacts are the contract between stages.
7) An initial application scaffold referenced a nonexistent model file and a mismatched schema (including a wrong yield unit); it was rebuilt to match the final pipeline.

TABLE XIII

GROUP CONTRIBUTION RECORD

Stage	Author(s)
Data sourcing, merging, cleaning, initial data dictionary	Nico Andrew O. Gallardo
EDA, split, preprocessing, augmentation, training, validation metrics	Charles John R. Cabatian
Test evaluation, final model selection, deployment	Renz Jerik C. Viloria

APPENDIX F
GROUP CONTRIBUTION RECORD

APPENDIX G
OWNERSHIP AND AUTHORSHIP DECLARATION

We, the undersigned, declare that this paper is the original work of the authors named above; that all borrowed datasets, methods, definitions, code, figures, and claims are acknowledged and cited in the reference list; that the data are public and no confidential data were used; and that every

reported number was checked against the project repository. AI assistants were used for planning, code drafting, and writing support; the authors reviewed the output and take full responsibility for the content.

Charles John R. Cabatian	Date
Nico Andrew O. Gallardo	Date
Renz Jerik C. Viloria	Date